

GUNNISON-CRESTED BUTTE, CO, USA

WMO: 724677

Lat: 38.533N

Lon: 106.933W

Elev: 2337

StdP: 76.23

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93007

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-23.8	-32.3	0.2	-24.5	-29.8	0.3	-23.0	8.8	-1.7	7.8	-1.4	0.5	70	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	28.1	12.0	27.4	11.7	26.2	11.4	14.2	22.7	13.6	22.1	13.0	21.5	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	11.6	16.2	11.0	10.9	16.0	9.8	10.0	16.1	47.9	23.2	46.1	21.1	44.2	21.3	16.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.5	8.3	7.3	DB	-30.7	30.5	3.1	1.3	-33.0	31.4	-34.8	32.1	-36.6	32.8	-38.9	33.7	(4)
(5)			WB	-30.1	15.2	2.7	0.9	-32.1	15.9	-33.7	16.4	-35.2	16.9	-37.2	17.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-11.2	-8.5	-1.1	4.2	9.1	14.2	17.6	16.1	11.9	5.5	-1.6	-9.5	(6)
	DBStd	10.61	5.66	5.68	4.90	3.25	3.27	2.52	1.78	1.74	2.94	3.46	4.58	6.08	(7)
	HDD10.0	2857	656	517	344	174	53	3	0	0	15	142	348	604	(8)
	HDD18.3	5258	914	751	602	423	285	125	36	70	194	397	598	862	(9)
	CDD10.0	656	0	0	0	1	27	129	234	191	71	3	0	0	(10)
	CDD18.3	14	0	0	0	0	0	1	11	2	0	0	0	0	(11)
	CDH23.3	1053	0	0	0	0	19	269	478	233	52	1	0	0	(12)
	CDH26.7	166	0	0	0	0	2	37	98	25	4	0	0	0	(13)
(14) Wind	WSAvg	2.2	1.2	1.6	2.3	3.2	3.1	2.8	2.4	2.2	2.1	2.2	1.7	1.3	(14)
(15) Precipitation	PrecAvg	260	20	17	15	16	18	16	33	39	28	20	16	20	(15)
	PrecMax	418	61	62	84	58	42	63	97	92	84	105	48	79	(16)
	PrecMin	153	0	0	0	0	0	0	4	4	0	0	0	0	(17)
	PrecStd	58	15	12	15	12	12	13	20	20	18	18	11	17	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	7.4	17.2	19.9	26.1	29.2	30.9	28.5	27.3	22.2	15.6	7.8	(19)
		MCWB	0.2	1.1	4.9	6.5	9.8	11.8	12.5	12.4	11.8	8.3	4.9	1.2	(20)
	2%	DB	2.4	4.6	12.9	17.7	22.8	27.6	28.9	27.4	24.8	19.5	12.7	4.4	(21)
		MCWB	-1.0	-0.1	3.3	5.3	8.3	11.0	12.3	12.0	10.7	7.4	3.5	-0.4	(22)
	5%	DB	0.3	2.5	10.9	16.2	21.3	26.4	27.7	26.2	23.0	17.5	10.2	2.3	(23)
		MCWB	-2.1	-0.9	2.6	4.8	7.7	10.4	12.1	11.8	9.7	6.4	2.7	-1.3	(24)
	10%	DB	-2.0	0.9	7.7	13.8	19.2	24.9	26.4	24.1	22.0	15.5	7.4	0.0	(25)
		MCWB	-4.0	-1.8	1.3	3.8	7.0	9.8	12.2	11.8	9.6	5.6	1.0	-2.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.9	5.4	7.2	10.5	13.4	15.4	15.1	13.3	10.0	5.6	2.0	(27)
		MCDB	4.1	5.0	14.9	17.6	23.6	25.3	24.8	23.7	22.7	17.4	13.6	6.4	(28)
	2%	WB	-0.5	0.6	4.0	6.0	9.2	12.3	14.6	14.1	12.3	8.4	4.1	0.4	(29)
		MCDB	1.9	3.9	12.3	15.8	20.6	23.8	23.5	21.8	20.7	17.7	11.3	3.7	(30)
	5%	WB	-2.1	-0.6	2.8	5.2	8.4	11.4	14.0	13.5	11.4	7.2	2.8	-1.0	(31)
		MCDB	0.0	2.5	9.3	14.6	19.1	23.0	22.6	21.4	19.6	15.9	9.7	1.9	(32)
	10%	WB	-3.8	-2.0	1.6	4.2	7.5	10.7	13.4	12.8	10.5	6.1	1.4	-2.6	(33)
		MCDB	-1.8	0.5	7.4	13.1	18.2	22.4	21.9	20.9	18.7	14.3	7.0	-0.1	(34)
(35) Mean Daily Temperature Range	MDBR		14.1	13.9	14.3	15.4	17.1	19.4	17.7	17.2	18.5	17.9	15.8	13.9	(35)
	5% DB	MCDBR	14.3	14.4	18.5	20.7	21.3	22.2	20.3	20.1	21.7	22.4	20.7	14.9	(36)
		MCWBR	11.0	10.4	10.1	10.3	9.5	9.1	7.7	7.8	9.7	11.9	12.4	10.5	(37)
	5% WB	MCDBR	13.3	13.5	16.8	18.8	18.9	19.2	15.8	15.8	16.9	19.3	19.3	13.6	(38)
		MCWBR	10.5	10.0	9.5	9.5	8.5	7.9	6.1	6.4	7.6	10.2	11.8	9.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.223	0.228	0.256	0.287	0.300	0.298	0.334	0.331	0.286	0.261	0.238	0.225	(40)
	taud		2.521	2.529	2.511	2.475	2.512	2.520	2.468	2.455	2.601	2.634	2.604	2.531	(41)
	Ebn at Noon		992	1026	1016	992	978	975	937	934	970	973	967	965	(42)
	Edh at Noon		73	84	96	107	105	104	109	108	87	76	68	68	(43)
(44) All-Sky Solar Radiation	RadAvg		2.67	3.51	4.95	5.95	6.62	7.55	6.59	5.88	5.30	4.05	2.94	2.31	(44)
	RadStd		0.16	0.22	0.39	0.35	0.51	0.47	0.38	0.27	0.32	0.31	0.20	0.17	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+12

Nomenclature: See separate page